

Spatial confounding and misaligned covariates in SDALGCP2

Restricted spatial regression, and covariates measured on a different support from the outcome

SDALGCP2 development notes

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Abstract

Two practical problems in areal disease mapping with spatially structured predictors. First, a spatially smooth covariate is collinear with the spatial random effect, which absorbs its signal and biases the regression coefficient (*spatial confounding*); we describe the restricted spatial regression implemented in SDALGCP2. Second, the covariate may be observed on a *different spatial support* from the outcome – other polygons, point monitors, or a coarser grid; we derive a change-of-support treatment that predicts the covariate to the candidate points feeding the intensity-scale offset and propagates the prediction uncertainty through a Berkson correction.

1 Background: the SDA-LGCP model

For aggregated counts Y_i over regions A_i , $i = 1, \dots, N$, the SDA-LGCP sets

$$Y_i \mid \boldsymbol{\eta} \sim \text{Poisson}(m_i e^{\eta_i}), \quad \boldsymbol{\eta} = \mathbf{D}\boldsymbol{\beta} + \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{R}(\phi)), \quad (1)$$

where $\mathbf{D}$ ($N \times p$) is the region-level design, m_i an offset, and $\mathbf{R}(\phi)$ the aggregated spatial correlation built from candidate points inside each region.

2 Part I: spatial confounding

2.1 The problem

When a covariate column of $\mathbf{D}$ is spatially smooth, it lies close to the span of the leading modes of $\mathbf{R}(\phi)$. The random effect $\mathbf{Z}$ can then reproduce much of that column, so the data are equally well explained by “covariate effect” or by “spatial noise”. The fitted $\hat{\boldsymbol{\beta}}$ is pulled away from the value a non-spatial GLM would give, usually *attenuated* toward zero, with inflated variance. In SDALGCP2 on a simulated gradient covariate (true $\beta = 0.5$) the standard fit returns $\hat{\beta} \approx 0.28$ while a Poisson GLM returns ≈ 0.42 – the spatial term has eaten the signal.

2.2 Restricted spatial regression (RSR)

Following Reich, Hodges & Zadnik (2006) and Hughes & Haran (2013), constrain the random effect to the orthogonal complement of the fixed-effect design. Let $\mathbf{K}$ ($N \times r$, $r = N - p$) be an orthonormal basis of $\text{null}(\mathbf{D}^\top)$, so

$$\mathbf{K}^\top \mathbf{K} = \mathbf{I}_r, \quad \mathbf{D}^\top \mathbf{K} = \mathbf{0}. \quad (2)$$

(SDALGCP2 takes $\mathbf{K}$ from the trailing columns of a complete QR of $\mathbf{D}$.) Replace the unrestricted random effect by $\mathbf{K}\alpha$:

$$Y_i \mid \alpha \sim \text{Poisson}(m_i e^{\eta_i}), \quad \eta = \mathbf{D}\beta + \mathbf{K}\alpha, \quad \alpha \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{K}^\top \mathbf{R}(\phi) \mathbf{K}). \quad (3)$$

Because $\mathbf{K}\alpha \in \text{col}(\mathbf{D})^\perp$, the random effect *cannot* reproduce any column of $\mathbf{D}$; β is identified by the data, not by the spatial structure, and the spatial term only smooths the residual. RSR removes the confounding by construction; in the Gaussian case $\hat{\beta}$ equals the OLS estimate, and for the Poisson model it is close to the GLM estimate.

2.3 Estimation: Laplace-approximate marginal likelihood

Rather than Monte Carlo, (3) is fitted by integrating α out at its mode. With $\Sigma_\alpha(\phi) = \mathbf{K}^\top \mathbf{R}(\phi) \mathbf{K}$ and $g(\alpha) = \sum_i \{y_i \eta_i - m_i e^{\eta_i}\} - \frac{1}{2\sigma^2} \alpha^\top \Sigma_\alpha^{-1} \alpha$, let $\hat{\alpha} = \arg \max_\alpha g(\alpha)$ (a few Newton steps; the negative Hessian is $\mathbf{H} = \mathbf{K}^\top \text{diag}(m_i e^{\eta_i}) \mathbf{K} + \Sigma_\alpha^{-1}/\sigma^2$). The Laplace-approximate marginal log-likelihood is

$$\ell(\beta, \sigma^2, \phi) = \sum_i \{y_i \hat{\eta}_i - m_i e^{\hat{\eta}_i}\} - \frac{1}{2} \left(r \log \sigma^2 + \log |\Sigma_\alpha| + \frac{\hat{\alpha}^\top \Sigma_\alpha^{-1} \hat{\alpha}}{\sigma^2} \right) - \frac{1}{2} \log |\mathbf{H}| + \text{const}, \quad (4)$$

maximised over $(\beta, \log \sigma^2)$ for each ϕ on a grid, with standard errors from the Hessian of (4). On the simulation above RSR returns $\hat{\beta} \approx 0.44$ (SE 0.07) – de-confounded – while still estimating the spatial parameters. Use `sda1gcp(..., control = sda1gcp_control(confounding = "restricted"))`.

3 Part II: covariates on a different support

3.1 The problem

The intensity-scale treatment of a continuous covariate needs its value $z(x_{ik})$ at the candidate points x_{ik} inside each outcome region, entering the offset $b_i(\beta) = \log \sum_k w_{ik} \exp\{z(x_{ik})^\top \beta\}$ (companion note on raster covariates). A raster gives z everywhere, so this is immediate. But often the covariate is observed on a *different support*:

1. **point** locations $s_1, \dots, s_L$ (air-quality monitors, weather stations);
2. **other polygons** $B_1, \dots, B_M$ (census tracts, catchments) that do not nest with the outcome regions A_i ;
3. a **coarser/finer grid** than the candidate points.

All three are *spatial misalignment* (change-of-support) problems (Gotway & Young, 2002; Gelfand, 2010): we must transfer z to the points x_{ik} .

3.2 Predict the covariate, then aggregate

Model the covariate as a latent spatial process $z(\cdot)$ and predict it to the candidate points from the observed data $\mathbf{z}^{\text{obs}}$.

Point support. Fit a geostatistical (Gaussian-process) model to $\{z(s_l)\}$ and kriging to each x_{ik} , giving a predictive mean and variance

$$\mu_z(x_{ik}) = \mathbb{E}[z(x_{ik}) \mid \mathbf{z}^{\text{obs}}], \quad v_z(x_{ik}) = \text{Var}[z(x_{ik}) \mid \mathbf{z}^{\text{obs}}]. \quad (5)$$

Areal support. If z is observed as averages over polygons B_j , $z_j^{\text{obs}} = |B_j|^{-1} \int_{B_j} z(u) du$, model the underlying surface $z(\cdot)$ and use *areal kriging*: the observations are linear (integral) functionals of z , so the joint Gaussian model gives the same predictive μ_z, v_z at the points x_{ik} – exactly the discrete-aggregation machinery SDALGCP2 already uses, applied to the covariate.

Grid support. A special case of point/areal kriging; bilinear interpolation is the cheap plug-in.

3.3 Berkson correction: propagate the prediction uncertainty

Plugging the kriged mean $\mu_z(x_{ik})$ straight into $b_i(\beta)$ ignores that $z(x_{ik})$ is *predicted*, not observed – a classic errors-in-variables problem that attenuates β . Because the kriging predictive distribution is Gaussian, $z(x_{ik}) \mid \mathbf{z}^{\text{obs}} \sim \mathcal{N}(\mu_z, v_z)$, we can integrate it out in closed form. Using $\mathbb{E}[e^{a^\top X}] = \exp(a^\top \mu + \frac{1}{2} a^\top V a)$ for $X \sim \mathcal{N}(\mu, V)$,

$$\mathbb{E}[\exp\{z(x_{ik})^\top \beta\} \mid \mathbf{z}^{\text{obs}}] = \exp\left\{\mu_z(x_{ik})^\top \beta + \frac{1}{2} \beta^\top V_z(x_{ik}) \beta\right\}, \quad (6)$$

so the offset that correctly accounts for the misalignment uncertainty is

$$b_i(\beta) = \log \sum_k w_{ik} \exp\left\{\mu_z(x_{ik})^\top \beta + \frac{1}{2} \beta^\top V_z(x_{ik}) \beta\right\}. \quad (7)$$

Equation (7) is the ordinary intensity-scale offset evaluated at the predicted covariate mean, *plus* a $\frac{1}{2} \beta^\top V_z \beta$ term that inflates the contribution of poorly-observed locations – the Berkson correction. It reduces to the raster case when $V_z \rightarrow 0$ (the covariate is known), and it is fitted by the same Gauss–Newton scheme as the raster model, with the effective covariate gradient $\partial b_i / \partial \beta$ now including the $V_z \beta$ term. Full propagation of the covariate model’s own parameter uncertainty (a two-stage or joint model) is a natural further step.

Summary. Misaligned covariates are handled in two steps that reuse the existing machinery: (i) predict z to the candidate points by (areal) kriging, obtaining μ_z, v_z ; (ii) feed them into the Berkson-corrected offset (7). The candidate-point grid that already discretises the outcome regions doubles as the common support on which outcome and covariate are aligned.

References. Reich, B.J., Hodges, J.S., Zadnik, V. (2006). Effects of residual smoothing on the posterior of the fixed effects in disease-mapping models. *Biometrics* 62, 1197–1206.

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